ST501 Final Project

Due: **12/7/23, 3pm, EST**

Instructions:

- Team formation:
  - Teams must consist of 3-5 members.
  - Designate one person as the primary contact.
  - The primary contact must email me the list of team members (names and emails) by **11/21/23, 3pm, EST**.
  - The primary contact person must upload the final report on Moodle.
- Final report must include:
  - Cover page that includes names of team members.
  - Itemized answers to all of the questions, which must be typed.
  - R files that were used to produce the results.
- Grading:
  - The code must run completely on my computer with no errors for full credit.
  - All plots and tables should look nice with appropriate titles and labels.

Problem #	**1**	**2**	**3**	**Total**
Total Points	20	40	40	
Points Earned				

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1. *How many observations do you need to cover an unknown percentile of a distribution?*

Suppose $X_1, \dots, X_n$ are iid random samples from some distribution F. We want to estimate the p-th quantile $Q_p = F^{-1}(p)$ for some $p \in (0,1)$. Determine the minimal sample size n needed to cover the p-th quantile with probability at least α, i.e. find the smallest n such that

$$P\left(X_{(1)} \leq Q_p \leq X_{(n)}\right) \geq \alpha.$$

a. Show that the minimal sample size n depends only on p (and not on F).
b. Find the minimum value of n for $p = 0.01, 0.05, 0.1$, and 0.2 and $\alpha = 0.95, 0.99$, and 0.999.

	$p = 0.01$	$p = 0.05$	$p = 0.1$	$p = 0.2$
$\alpha = 0.95$				
$\alpha = 0.99$				
$\alpha = 0.999$				

Note: Include the R code you use to fill out the above table.

2. *Visualizing convergence.*

   a. Let $X \sim \text{Exp}(\lambda = 1)$.

      i. Show that the minimum order statistic converges to 0 as $n \to \infty$.

      ii. For sample sizes of $n = 1, 2, \ldots, 50$, generate $N = 1000$ datasets of $X_1, \ldots, X_n \sim \text{Exp}(\lambda = 1)$.

         1. Create a line plot with the sample size n on the x-axis and the minimum value on the y-axis.
         2. For $\varepsilon = 0.05$, approximate $P(|X_{(1)} - 0| < \varepsilon)$ using each of the simulated minimum values from the $N = 1000$ generated datasets above.
         3. Create a line plot with the sample size n on the x-axis and the estimated probability on the y-axis.
         4. How does plot demonstrate convergence in probability?

   b. Let $Y \sim \text{Pois}(\lambda)$.

      i. For sample sizes of $n = 5, 10$, and 100, and $\lambda = 1, 5$, and 25, generate $N = 1000$ datasets of $Y_1, \ldots, Y_n \sim \text{Pois}(\lambda)$.

         1. For each n and λ, create a histogram of the sample means from the corresponding generated dataset.
            a. Each bin of the histogram should contain a single value, eg with $n = 5$, the bins should be $0, 0.2, 0.4, 0.6, \ldots$
            b. Overlay the appropriate limiting normal distribution over the histogram.
         2. For each n and λ, use the $N = 1000$ sample means to approximate $P(\bar{Y} \geq \lambda + 2 \cdot \sqrt{\lambda/n})$ and compare to the probability approximated by the normal distribution.
         3. How do these plots demonstrate convergence in distribution?

Note: For (b.i.1), arrange the 9 plots in a 3x3 grid with n increasing from left to right and λ increasing from top to bottom.

Note: For (b.i.2), provide your answers in two tables (one for the Monte Carlo approximation and one for the normal approximation) as follows:

	$n = 5$	$n = 10$	$n = 100$
$\lambda = 1$			
$\lambda = 5$			
$\lambda = 25$			

3. Let X_1 and X_2 be random variables with $E[X_i] = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2$ for $i = 1, 2$. WLOG, assume $\sigma_2 \geq \sigma_1 > 0$. Let ρ be the correlation between X_1 and X_2.

   Define the weighted random variable $X(w) = wX_1 + (1-w)X_2$ for $w \in [0,1]$.

   a. For any $w \in [0,1]$, find c_0, c_1, a, b, and c such that

   $$E[X(w)] = \mu(w) = c_0 + c_1 w$$
   $$\text{Var}\big(X(w)\big) = \sigma^2(w) = aw^2 - 2bw + c$$

   b. Suppose we want to find $w \in [0,1]$ that maximizes $\mu(w)$ subject to the constraint that $\sigma^2(w) \leq \sigma_1^2$.
      i. If $\mu_1 \geq \mu_2$, show that w = 1 solves the optimization.
      ii. If $\mu_1 < \mu_2$, show that the optimal w is given by the smallest $w \in [0,1]$ that satisfies $\sigma^2(w) \leq \sigma_1^2$.

   c. We can use X_1 and X_2 to model two stocks in a portfolio (see below).
      i. Estimate c_0, c_1, a, b, and c using the moment estimates of the AAPL and MSFT return prices.
      ii. Estimate the optimal w according to (b).
      iii. Estimate the value-at-risk ($V_a R$) separately for both AAPL and MSFT with confidence $\alpha = 0.95$.
      iv. Find w that minimizes the $V_a R$ for the combined portfolio of AAPL and MSFT for $\alpha = 0.95$.

```
# install.packages("quantmod")
library(quantmod)

getSymbols(c("AAPL","MSFT"),from = "2018-10-31", to = "2023-11-01")

# daily return values
AAPL.returns <- as.numeric(dailyReturn(AAPL))
MSFT.returns <- as.numeric(dailyReturn(MSFT))

# trading days
days <- as.Date(.indexday(AAPL))

# plot
plot(days, AAPL.returns, typ = "l", ylab = "Returns", col = "red")
lines(days, MSFT.returns, col = "blue", lwd = 1.5)
legend("topright", legend = c("AAPL", "MSFT"), col = c("red", "blue"), lty =
1)

# summary
summary(cbind(AAPL.returns, MSFT.returns))
cov(cbind(AAPL.returns, MSFT.returns))
```